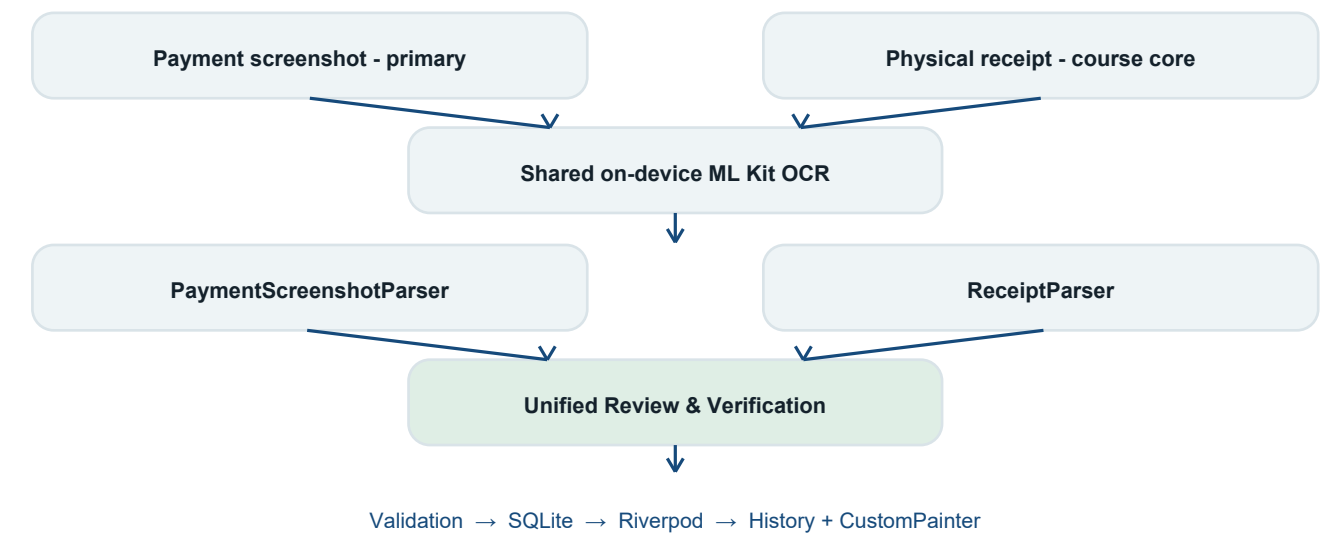


Smart Expense Capture & Tracker

Technical report · Mini-Project 3 · Flutter / Dart · October 2026

Vietnamese users often keep a bank or e-wallet payment screenshot rather than a paper receipt. The app therefore offers screenshot import as its primary entry, while retaining the course-required physical receipt camera and OCR workflow. Both inputs use on-device recognition and require human verification before local storage.

Architecture



Course core and product extension

Course required core	Product extension
Camera/gallery receipt, ML Kit OCR, receipt total/date/merchant heuristics, review, SQLite CRUD, Riverpod 2, custom charts.	Bank and e-wallet screenshot import, payment-specific parser, status, source label, optional provider/reference, duplicate warning.

Design: Material 3, a restrained blue/green palette, solid surfaces, 48 dp touch targets, light/dark themes, an original receipt/check brand mark, Vietnamese-first and English UI, and one review form adapted to the input source.

OCR heuristics and persistence

Recognition uses `google_mlkit_text_recognition` with the Latin model. The parsers apply Dart RegExp and ordered rules; uncertain fields stay blank. No OCR value is committed directly.

Field	Detection rule	Review fallback
Receipt total	Weighted total/tổng tiền/thanh toán/cộng tiền/amount due line; adjacent amount line; VND grouping.	Blank amount
Receipt merchant	First plausible header; reject invoice, address, phone and date labels.	Blank merchant
Date/time	Strict dd/MM/yyyy or yyyy-MM-dd, plus HH:mm for payments; reject impossible dates.	Editable date/time
Payment amount	Prefer số tiền/amount and currency-bearing lines; normalize 150000, 150.000, 150,000.	Blank amount
Recipient/status	Labelled recipient or cautious post-date name; detect failed/pending before success phrases.	Edit recipient; require success
Transfer note	Labelled content, transfer phrase, or text after a destination account number; never copy the number.	Optional editable note
Duplicate	Reference match, or same source + amount + date + recipient.	Warn; user may save anyway

Data and state

SQLite stores integer VND amounts, date/time, category, source, merchant/recipient, optional status/provider/reference/note, image path, and timestamps. Version 2 adds payment columns with an additive migration. ExpenseDatabase owns SQL; a Riverpod 2 AsyncNotifier refreshes history and analytics after each successful mutation.

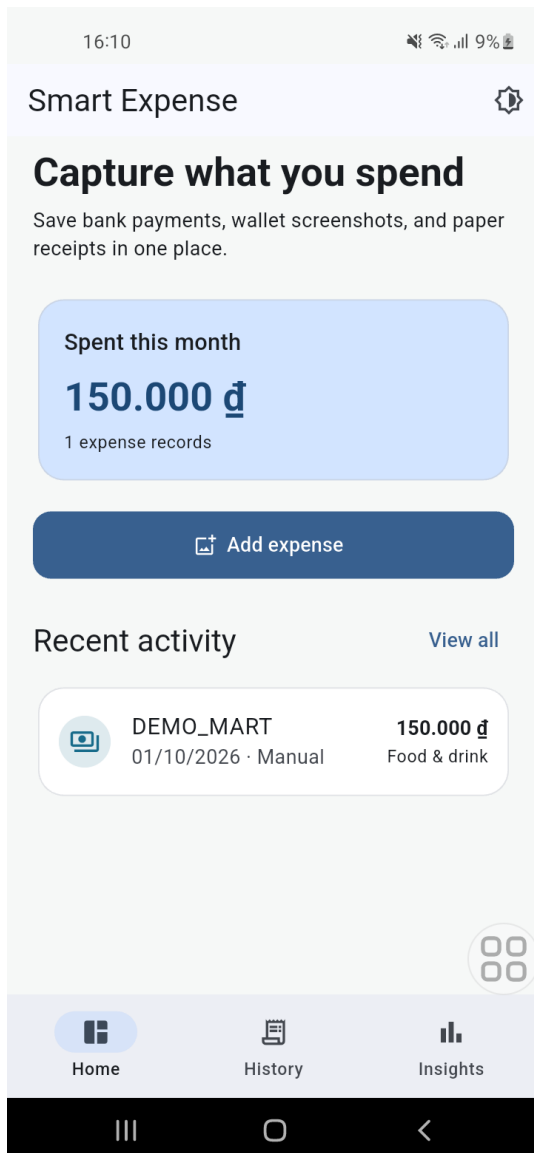
Privacy, testing, and limits

OCR and image processing stay on-device. Payment OCR text is visible during review but not persisted because it may contain account numbers; screenshots remain in app-owned storage. Unit/widget tests cover receipt and payment parsing, form correction/validation, SQLite CRUD, duplicate detection, and aggregation. The Android integration test mocks picker/ML Kit while exercising the real parsers and UI.

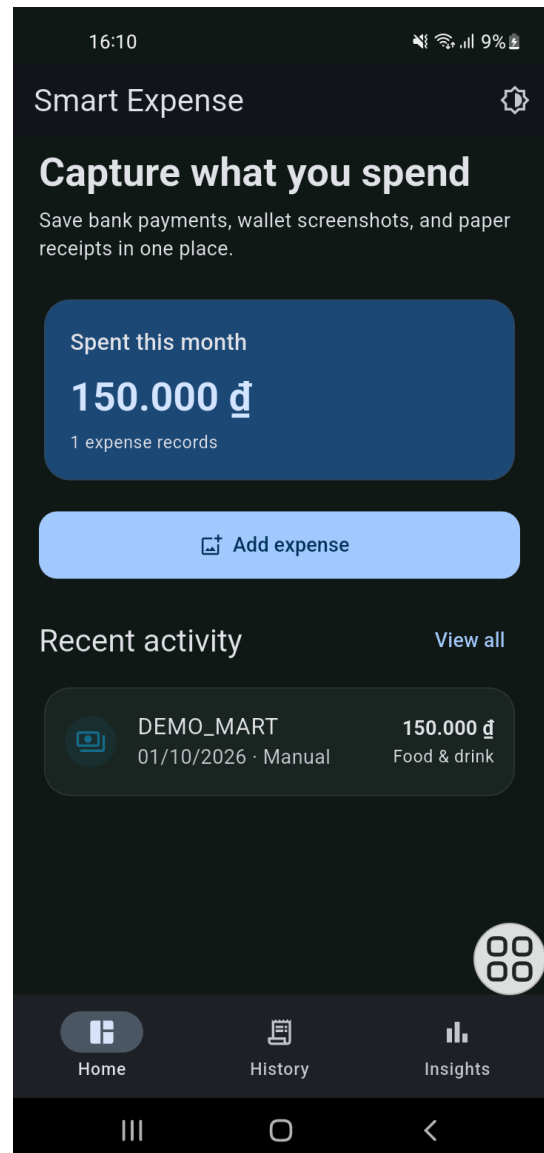
Limitations: noisy or handwritten inputs need correction; ambiguous VND decimals are not guessed; bank layouts vary; duplicate matching is advisory. Release OCR was exercised with three payment screenshots on a Samsung Android phone; camera receipt quality still needs a real paper receipt. iOS build validation requires macOS and Xcode.

Material 3 interface

Screenshots below were captured from the Android build before the Vietnamese labels and launcher icon were added. The same navigation and review hierarchy remains in the current release.



Light mode



Dark mode

Demo sequence

Open app → scan a real receipt → show OCR text and Regex-derived fields → correct in Review & Verification → save → history → animated category donut and weekly bars. Then import a bank or wallet screenshot, verify status/recipient/amount, and show the combined analytics update.

Implementation references: repository PRD and README; Flutter documentation; google_mlkit_text_recognition and image_picker package documentation on pub.dev.